Questions for Advisors

Meeting preparation — Professor Zhu & Wenbin

Nick Mino | 29 July 2026

Top three (please prioritize)

1. Is the current optimization-selected UQ formulation sufficiently close to the intended research direction?
2. Is the Part I / Part II decomposition the right project structure?
3. Should the first theorem focus on one motivating constrained-optimization instance before a general result?

A. Mathematical formulation

1. Is the current uncertainty-quantification formulation now sufficiently close to the intended research direction?
2. Is the Part I / Part II decomposition the right way to structure the project?
3. Is replace-one stability the appropriate stability notion for the optimizer in this setting?
4. Are the proposed assumptions on the objective, constraints, feasible set, regularity, and perturbation mechanism reasonable (as sketched in the problem write-up)?
5. Should the first theorem focus on one motivating constrained-optimization problem before pursuing a general result?

B. Novelty and literature positioning

1. Does the intended contribution appear sufficiently distinct from existing conformal risk control, decision-focused uncertainty quantification, and post-selection inference work?
2. Which papers should be treated as the closest technical baselines?
3. Should Part II aim first for a direct coverage bound, a risk-control statement, or a more general transfer theorem from stability to validity?

C. Research sequencing

1. Should further operator-library expansion stop until Part I is complete?
2. Which concrete special case of constrained optimization would be most useful for validating the general theorem?
3. What result would constitute meaningful progress before the next meeting?

D. Continued collaboration

1. What meeting cadence is realistic for the remainder of SURA?
2. Is continuing this line after SURA appropriate if Part I lands cleanly?
3. What milestones would make the work suitable for a paper or longer-term collaboration?

Context documents: `Research_Update_Summary.pdf` (overview); `Formulation/writeup/Problem_Writeup.pdf` (mathematics); `Intellectual_Timeline.pdf` (history).